

AI-Shield Moderation — Unit Test Report

Automated test execution report generated via Jest + ts-jest

Project	ai-shiled-moderation
Test Runner	Jest 30.4.2 (ts-jest, Node/TypeScript)
Report Generated	2026-07-09
Total Duration	~11.9s
Result	SUCCESS — All tests passed

11
TEST SUITES

76
TOTAL TESTS

76
PASSED

0
FAILED

Test Suite Results

Test Case	Status	Duration
PiiDetector — src/detectors/pii.detector.test.ts (11 tests)		
allows text with no PII	PASS	2 ms
detects an email address and returns a MASK action	PASS	1 ms
detects an Indian phone number	PASS	1 ms
reports position start/end for each finding	PASS	<1 ms
collects multiple findings across rules	PASS	<1 ms
ignores disabled rules	PASS	1 ms
treats a missing text field as empty string	PASS	<1 ms
returns ALLOW immediately when the detector is disabled in config	PASS	2 ms
mask() replaces detected PII with the configured mask token	PASS	<1 ms
mask() returns the original text unchanged when nothing matches	PASS	<1 ms
mask() masks multiple occurrences of the same rule	PASS	1 ms
CiiDetector — src/detectors/cii.detector.test.ts (7 tests)		
allows text with no confidential info	PASS	3 ms
detects a keyword match case-insensitively	PASS	2 ms
detects a pattern match with position info	PASS	1 ms
ignores disabled rules	PASS	<1 ms
returns ALLOW when detector disabled in config	PASS	1 ms
mask() redacts keyword matches with the type-derived token	PASS	1 ms
mask() redacts pattern matches with the type-derived token	PASS	<1 ms
SecretDetector — src/detectors/secret.detector.test.ts (6 tests)		
allows text with no secrets	PASS	2 ms
blocks when an API key pattern is found	PASS	1 ms

partially masks the raw value in findings (first 6 chars + ****)	PASS	1 ms
detects AWS keys independently of openai keys	PASS	<1 ms
returns ALLOW when detector disabled in config	PASS	1 ms
treats missing text as empty and allows	PASS	<1 ms
ToxicDetector — src/detectors/toxic.detector.test.ts (7 tests)		
allows clean text	PASS	3 ms
blocks high severity toxic content	PASS	2 ms
blocks medium severity when configured in blockOnSeverity	PASS	<1 ms
allows low severity content that is not in blockOnSeverity	PASS	<1 ms
matches keywords case-insensitively	PASS	<1 ms
ignores disabled categories	PASS	1 ms
returns ALLOW when detector disabled in config	PASS	<1 ms
InjectionDetector — src/detectors/injection.detector.test.ts (6 tests)		
allows normal text	PASS	4 ms
blocks a known injection phrase	PASS	1 ms
matches patterns case-insensitively	PASS	1 ms
collects findings from multiple rule types	PASS	<1 ms
returns ALLOW when detector disabled in config	PASS	1 ms
treats missing text as empty and allows	PASS	<1 ms
FileDetector — src/detectors/file.detector.test.ts (6 tests)		
allows a valid, in-limit file	PASS	5 ms
returns ALLOW with a message when no file is provided	PASS	1 ms
blocks a disallowed file extension	PASS	<1 ms
blocks a file exceeding the max size	PASS	<1 ms
accepts a file exactly at the max size boundary	PASS	1 ms
returns ALLOW when detector disabled in config	PASS	1 ms
TokenDetector — src/detectors/token.detector.test.ts (6 tests)		
allows text within the token limit for a known model	PASS	2 ms
blocks text exceeding the token limit for a known model	PASS	<1 ms
falls back to the "default" model rule when model is unspecified	PASS	1 ms
falls back to the "default" model rule for an unrecognized model name	PASS	1 ms
estimates tokens by ceiling division of text length by charsPerToken	PASS	<1 ms
returns ALLOW when detector disabled in config	PASS	1 ms
DecisionEngine — src/engine/decision.engine.test.ts (14 tests)		
resolveAction › returns ALLOW when there are no results	PASS	6 ms
resolveAction › returns ALLOW when all results allow	PASS	1 ms
resolveAction › returns MASK when any result is MASK and none are BLOCK	PASS	1 ms
resolveAction › prioritizes BLOCK over MASK and ALLOW	PASS	1 ms

applySanitization › returns text unchanged when there are no MASK results	PASS	1 ms
applySanitization › applies PII masking when the pii detector result is MASK and triggered	PASS	2 ms
applySanitization › applies CII masking when the cii detector result is MASK and triggered	PASS	1 ms
applySanitization › skips masking for a MASK result that is not triggered	PASS	<1 ms
applySanitization › applies both pii and cii masking together	PASS	1 ms
buildResponse › omits sanitizedContent when the final action is BLOCK	PASS	3 ms
buildResponse › includes sanitized content when the final action is MASK	PASS	1 ms
buildResponse › computes metadata: tokenEstimate, processingTimeMs, timestamp	PASS	1 ms
buildResponse › merges extra.metadata into the response metadata	PASS	1 ms
buildResponse › always sets status to success	PASS	1 ms
errorMiddleware — src/middleware/error.middleware.test.ts (6 tests)		
logs the error message and stack	PASS	11 ms
responds with a 500 status	PASS	1 ms
responds with a BLOCK action and the error message in the body	PASS	4 ms
falls back to a generic message when err.message is empty	PASS	2 ms
includes metadata with a zeroed tokenEstimate/processingTimeMs and an ISO timestamp	PASS	2 ms
does not call next()	PASS	1 ms
requestLogger — src/middleware/request.logger.test.ts (3 tests)		
logs method, path, and content-type from the request	PASS	2 ms
calls next() exactly once	PASS	1 ms
handles a missing content-type header gracefully	PASS	1 ms
loadRules — src/config/loader.test.ts (4 tests)		
reads and parses the JSON rules file	PASS	3 ms
caches the parsed result and does not re-read the file when mtime is unchanged	PASS	1 ms
re-reads the file when mtime changes	PASS	1 ms
maintains independent cache entries per filename	PASS	1 ms

Code Coverage Summary

Coverage is high (mostly 90–100%) for the modules under test — detectors, the decision engine, middleware, and the rules loader. Express route handlers (src/routes/**), app.ts, and utils/logger.ts show 0% because no integration tests exist for them yet; the unit tests in this run only exercise business logic in isolation via mocked dependencies.

File / Directory	% Stmts	% Branch	% Funcs	% Lines	Uncovered Lines
All files	47.34	65.06	75.00	44.61	
src	0	100	100	0	
app.ts	0	100	100	0	1-18

src/config	100	100	100	100	
loader.ts	100	100	100	100	
src/detectors	99.33	93.75	100	100	
base.detector.ts	100	100	100	100	
cii.detector.ts	100	95	100	100	14
file.detector.ts	100	93.75	100	100	20
injection.detector.ts	100	100	100	100	
pii.detector.ts	100	100	100	100	
secret.detector.ts	94.11	90	100	100	18
token.detector.ts	100	92.85	100	100	14
toxic.detector.ts	100	87.5	100	100	14,33
src/engine	100	100	100	100	
decision.engine.ts	100	100	100	100	
src/middleware	100	100	100	100	
error.middleware.ts	100	100	100	100	
request.logger.ts	100	100	100	100	
src/routes (+ detect/*)	0	0	0	0	
index.ts, moderate.route.ts, detect/*.route.ts (8 files)	0	0	0	0	no tests
src/utils	0	0	100	0	
logger.ts	0	0	100	0	1-30